Anatomy-Aware Pose Estimation on Lightweight Networks

Abstract: As robots increasingly operate in human-centric environments, ensuring safe boundaries becomes a critical requirement. This requires eliminating any anatomically inconsistent predictions under human body occlusion, which significantly increases risk in Human-Robot Interaction: standard models often predict dislocated or unrealistic joints and limb lengths.

This project addresses the challenge of anatomically consistent human pose estimation under occlusion, enforcing explicit geometric constraints deployed on a lightweight framework capable of real-time inference on edge devices - bridging the gap between theoretical biomechanical priors and safe, practical deployment in robotic environments.

Dataset: MS-COCO Keypoints dataset and OCHuman dataset

Task: Human pose estimation under occlusion remains an open challenge, as current models frequently produce anatomically inconsistent predictions when keypoints are partially or fully occluded. Pytel et al. (ICPR 2021) demonstrate that occlusion-specific data augmentation does not enforce top-down models' robustness in either HRNet or SimpleBaseline. It even degrades performance on bottom-up approaches such as Higher HRNet. On the other hand, Han et al (2025) address occluded pose estimation through two complementary strategies. Firstly, they implement a limb joint occlusion augmentation approach, that simulates real-life occlusion during training, improving the generalization capabilities of the model; secondly, they develop a Dynamic Structure Loss based on limb graphs that models statistical dependencies between adjacent joints. Similarly, Liu et al. (DSP 2024) introduce constraint-based losses on bone lengths and keypoint counts, demonstrating improvements in structural consistency. However, joint angle constraints are not taken into account in either formulation, representing a primary source of anatomically impossible configurations under occlusion. Xu et al. (NeurIPS 2022) show that a plain ViT backbone with a lightweight decoder achieves state-of-the-art performance while remaining architecturally scalable for real-time deployment on edge hardware. Finally, Zhang et al. (CVPR 2019) demonstrate that tasks depending on skeletal topology fail when pose predictions are anatomically incoherent, establishing the motivation of our work in human-robot interaction scenarios.

To the best of our knowledge, no existing work addresses both geometric ordering of skeletal landmarks and anatomical constraints on a lightweight framework validated for real-time HRI deployment. We propose to fill this gap, combining the Skeletal Topology Loss, a training objective that combines bone length, joint angles, and geometric ordering constraints, deployed on an efficient backbone suitable for edge hardware inference. The experiment also investigates the role of occlusion-specific data augmentation, reconciling the contrasting findings of Pytel et al., who show its ineffectiveness, and Han et al., who demonstrate improvements through structured limb joint augmentation.

The process would be visible thanks to a Grad-CAM implementation, to focus on the visual attention on the model during the prediction phase, thanks to a spatial importance map. This would demonstrate that the kinematic constraints enforce reasoning over the full skeletal context.

All is concluded by a specific metric evaluation, called Anatomical Violation Rate (AVR), to quantify the structural plausibility of the predicted pose, by penalizing structural inconsistencies.

To assess the robustness of the proposed STL, we adopt a strict cross-dataset evaluation protocol. The model training is performed on the MS-COCO Keypoints dataset and evaluated directly on the OCHuman benchmark, with no fine-tuning, to isolate the generalization capability of the learned anatomical priors. It would demonstrate the ability to generalize and correctly predict any possible situation in a real world environment.

Main objectives:

- *Skeletal Topology Loss (STL) formulation:* Develop a multi-factor loss function integrating bone length, joint angle, and geometric ordering constraints (modeled as a kinematic tree) to prevent anatomically impossible configurations under severe occlusion.
- *Real-time edge deployment:* Validate anatomical constraints on a lightweight architecture, demonstrating that structural accuracy is achievable without heavy computation, ensuring suitability for low-latency human-robot interaction (HRI).
- *Anatomical Violation Rate (AVR) metric:* Introduce a safety-critical evaluation metric that, unlike standard mAP, directly quantifies the structural plausibility of predicted poses to assess the risk of incoherent predictions.
- *Ablation study:* Isolate the contribution of each STL component (bone ratio, joint angle, geometric ordering) and analyze the individual and combined impact on AP, AR, and AVR.
- *Explainable pose estimation:* Integrate Grad-CAM techniques to map the model's visual attention, verifying its ability to exploit the anatomical context of visible joints to infer occluded ones, thereby increasing decision transparency.
- *Cross-dataset robustness:* Evaluate generalization through a strict MS-COCO → OCHuman protocol without fine-tuning, demonstrating that the learned universal anatomical priors enable robust performance in unseen real-world occlusion scenarios.

References:

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